Scribble-Based Image Segmentation

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Summer 2025

1 Introduction

This project tackles binary image segmentation from sparse scribbles, evaluated using mIoU. We compare three paradigms: unsupervised clustering (Seeded K-Means), classical optimization (GrabCut), and deep learning (U-Net). Our findings show that a hybrid approach, refining the U-Net's output with GrabCut, yields the best performance by combining deep learning's semantic understanding with the local precision of classical methods.

2 Data analysis and reprocessing

The dataset consists of 228 images with sparse but highly reliable scribble annotations (mean coverage 1.6%, less than 0.1% disagreement with ground truth). This sparsity, combined with a significant object/background class imbalance (median object ratio of 0.157), motivated a dual strategy: using scribbles as both an input feature and a hard post-processing constraint, and a class-imbalance-aware loss function for our deep learning model.

To ensure a fair comparison, we applied a standardized pipeline to all models. All images and masks were first resized to 256x256 pixels, and the training set was expanded using geometric augmentations like random flips and rotations. The classical K-Means and GrabCut models were applied directly to these 3-channel RGB images. In contrast, the U-Net required a more extensive pipeline to maximize performance: its training data underwent heavy color and noise augmentations (random horizontal flips, rotations, scaling, translations, color jitter, mild Gaussian blur, unsharp masking, and occasional grayscale conversion), and its input was expanded to a 5-channel tensor by adding two channels derived from the dilated scribbles.

3 ML modeling

We implemented and evaluated three distinct models, each representing a different paradigm of machine learning. All models were developed in Python, and their key hyperparameters were tuned using a fixed validation set which is detailed in Section 4.

3.1 Unsupervised Baseline: Seeded K-Means

Our simplest baseline is a direct application of Unsupervised Learning (Clustering). Implemented with OpenCV's `cv2.kmeans` function, the algorithm first groups all image pixels into k color clusters[1]. The final segmentation is then formed by the union of all clusters that are "seeded" by the provided foreground scribbles.

3.2 Classical Optimization Model: GrabCut

As a more sophisticated classical method, we used GrabCut, a graph-based algorithm that draws on concepts from Optimization and Probabilistic Modeling[2]. Implemented with OpenCV's `cv2.grabCut`, it models the image as a graph and iteratively refines a segmentation by finding a minimum cut that separates the foreground and background color models while favoring smooth boundaries.

3.3 Deep Learning Model: U-Net

To represent the Deep Learning paradigm, we implemented a U-Net in PyTorch. Our architecture features a 4-block encoder and a 4-block decoder with skip connections. The fundamental building block is a DoubleConv unit (two consecutive 3x3 convolutions with Batch Normalization and ReLU)[3]. The model was customized to accept a 5-channel input, as mentioned in Section 2.

As established in our data analysis, the severe class imbalance in the dataset required a specialized loss function. To mitigate this, we trained the model to minimize a combined loss of Binary Cross-Entropy(CE) and Dice Loss, as Dice Loss is particularly effective for imbalanced segmentation tasks. The combined loss is $\mathcal{L}_{\text{combined}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} + 0.5 \times \mathcal{L}_{\text{scribble}}$. For final predictions, we improved robustness by employing Test-Time Augmentation, averaging the outputs from the original and a horizontally flipped image.

To combine the U-Net's semantic robustness with GrabCut's boundary precision, we further created a hybrid U-Net-GC model. The method works by using the U-Net's prediction to create a three-level map for GrabCut: a high-confidence core of the object from an eroded mask is treated as definite foreground, an uncertain boundary region, derived from dilation, is treated as probable foreground, and the rest as definite background. This allows GrabCut's optimization to focus solely on the uncertain border, snapping the final segmentation to the most robust pixel-level evidence in the image.

3.4 Hybrid Model: U-Net Guided GrabCut Refinement

4 Model selection

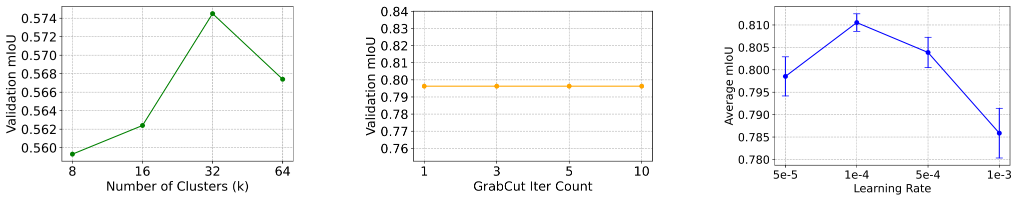
4.1 Evaluation Strategy

The 228 images in the training set were divided using a fixed 85/15 train/validation split. This fixed split ensures that all models and hyperparameter configurations were evaluated on the exact same data, allowing for a fair comparison. The metric for all evaluations was the mIoU on the validation set. For the U-Net model, performance was monitored at the end of each training epoch. We implemented early stopping by tracking the validation mIoU and saving the model checkpoint that achieved the highest score to prevents overfitting.

4.2 Hyperparameter Tuning and Final Selection

We performed a systematic search to find the optimal hyperparameters for each model. The configuration that yielded the highest validation mIoU is selected.

For Seeded K-Means, we tuned the number of clusters (`k`), finding that performance peaked at k=32 (mIoU=0.5745), as higher values led to over-segmentation. For GrabCut, tuning the number of iterations (`iterCount`) showed stable performance, likely because the algorithm converges quickly with the highly informative initial scribbles. For the U-Net, we tuned the `learning rate` and found the optimal validation mIoU of 0.8130 was achieved at 1e-4.



(a) K-means (b) GrabCut (c) U-Net

Figure 1: Hyperparameter tuning of three models

We select hybrid U-Net-GC as our final model for the challenge, for its superior performance and robustness. The U-Net used in this hybrid model was the optimal one mentioned above.

5 Empirical Results

5.1 Quantitative results

Table 1: Validation performance

Model	Background IoU	Object IoU	mIoU
Seed K-means	0.7207	0.4283	0.5745
GrabCut	0.9090	0.6835	0.7963
U-Net	0.9095	0.7165	0.8130
Hybrid Model(U-Net-GC)	0.9154	0.7452	**0.8303**

The final performance metrics are summarized in Table 1. The U-Net achieved the highest overall performance with an mIoU of 0.8130, followed by GrabCut, with the Seeded K-Means model as a simple baseline.

While the U-Net and GrabCut models performed similarly on the large background class, the U-Net's superior performance was driven by its significantly higher Object IoU, showing a better ability to identify the object of interest.

5.2 Qualitative Results

To provide visual context for these quantitative scores, Figure 2 shows the predictions of each model on a representative image from the validation set.

(a) Original (b) Scribble (c) Ground Truth

(d) K-means (e) GrabCut (f) U-Net (g) Hybrid

Figure 2: Qualitative comparison of model predictions. The first row shows the original image, user input, and ground truth. The second row shows predictions from different methods.

The qualitative results align with the metrics. The Seeded K-Means prediction is coarse and noisy, capturing only a rough outline of the object. GrabCut offers a substantial improvement, with a cleaner background and a more complete object shape. However, it may misclassify some edge areas. U-Net yields the most precise segmentation, preserving fine details in the object's contour. The hybrid model further refines U-Net's segmentation near the borders, achieving a result that most closely resembles the ground truth.

6 Discussion

6.1 Ablation Study: Justifying the U-Net's Design

To validate our two design choices for the U-Net - Heavy augmentation and the combined loss function, we performed an ablation study. The results provide two takeaways. First, comparing our combined CE+Dice loss (0.8026 mIoU) against a Dice-only loss (0.7993 mIoU) confirms that while Dice is crucial for handling class imbalance, the stable gradients from BCE are beneficial for achieving the best performance. Second, the significant performance drop when using only light augmentation (0.7801 mIoU) proves that the U-Net was data-starved and that the aggressive augmentation strategy was essential for robust generalization.

6.2 Robustness and the Motivation for a Hybrid Model

Table 2: Validation performance after data perturbations

Model	mIoU (Original)	mIoU (Noisy)	mIoU (Dark)
U-Net	0.8711	0.8054	0.7851
GrabCut	0.8041	0.6208	0.8001

To quantitatively assess model robustness, we conducted an experiment by applying two common perturbations—strong Gaussian noise and a drastic brightness reduction—to the validation set. The results, shown in Table 2, reveal that the models possess complementary strengths. The U-Net demonstrated strong general robustness, with its performance remaining stable—a direct result of its training on augmented data. In contrast, GrabCut's performance was brittle, failing when noise corrupted its color statistics, but it was resilient to uniform brightness changes.

This experiment reveals that the models possess complementary strengths: the U-Net provides a semantically robust initial prediction, while GrabCut offers a locally precise boundary refinement. This directly motivated our hybrid U-Net-GC model, which leverages the U-Net's high-level semantic prediction to guide GrabCut's pixel-perfect boundary optimization.

6.3 Explainability

The models present a classic trade-off in explainability. GrabCut is a model whose logic is based on interpretable color statistics. In contrast, the U-Net is a "black-box" whose complex, learned features are difficult to interpret. While this hybrid approach sacrifices the transparency of GrabCut, given that the primary objective of this challenge is to maximize mIoU, we still selected the Hybrid Model as our final approach, accepting the lack of explainability as a compromise.

7 Conclusion

Our investigation into scribble-based image segmentation showed that a hybrid U-Net-GrabCut model gave the best results. This approach succeeds by merging the semantic robustness of a deep learning model with the fine-grained boundary precision of a classical optimization algorithm.

References

[1] D. J. Bora and A. K. Gupta. Clustering approach towards image segmentation: An analytical study. *Int. J. Res. Comput. Appl. Robot.*, 2(7):115–124, 2014.

[2] S. Pang, T. H. G. Thio, F. L. Siaw, M. Chen, and Y. Xia. Research on improved image segmentation algorithm based on grabcut. *Electronics*, 13:4068, 2024.

[3] Y. Peng, M. Sonka, and D. Z. Chen. U-net v2: Rethinking the skip connections of u-net for medical image segmentation. *arXiv*, 2023.